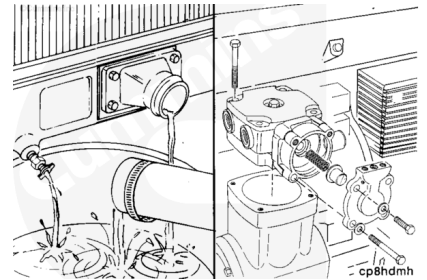


Trainings ▾

Disassemble

If the cylinder head is removed while the air compressor is on the engine, drain the engine coolant. Reference the specific engine application manual.

Since the valve plate, head and unloader body are indexed, marking of these parts is recommended to make sure they are reassembled in the proper orientation.



⚠ WARNING ⚠

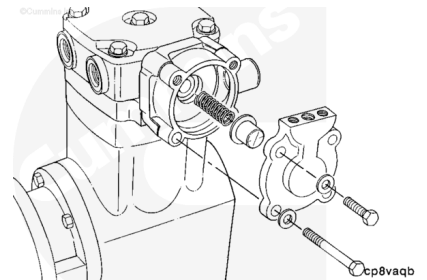
The unloader body is installed with spring tension. Use care when removing to prevent personal injury. Always wear protective eye wear.

Hold the unloader valve body down and remove the four capscrews.

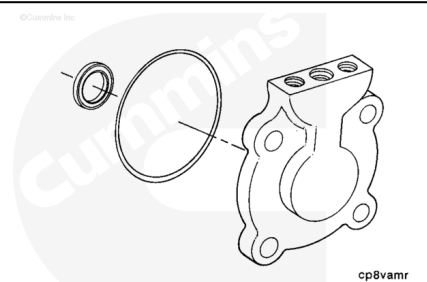
Remove the unloader valve body.

Remove the unloader valve spring.

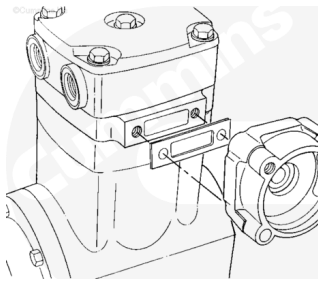
Remove the unloader valve cap.



Remove the unloader body gasket and unloader valve cap rectangular ring seal.



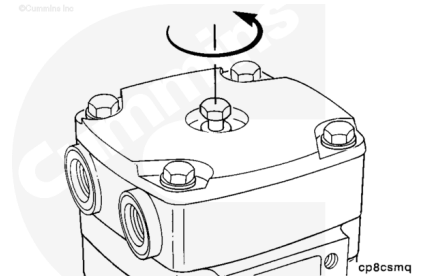
Remove the intake manifold and gasket.



cp8gkmf

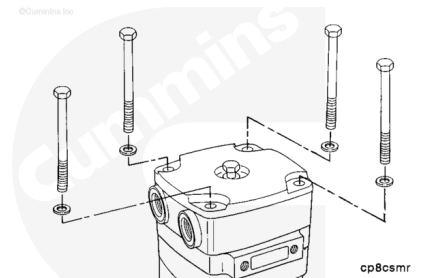
Loosen, but do **not** remove, the center head capscrew.

Mark the head for orientation during assembly.



cp8csmq

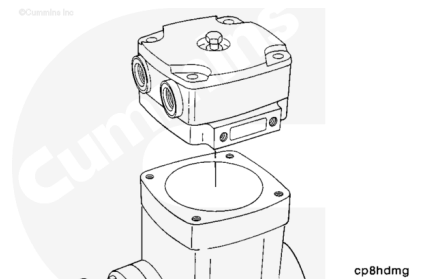
Loosen and remove the four corner head capscrews. Save and reuse the capscrews.



cp8csmr

Note : If continuing with disassembly of the head, valve plate, and cover, make sure the work surface is clean. Debris pushed into the valve sealing surfaces by setting components on a dirty surface will cause a malfunction after assembly.

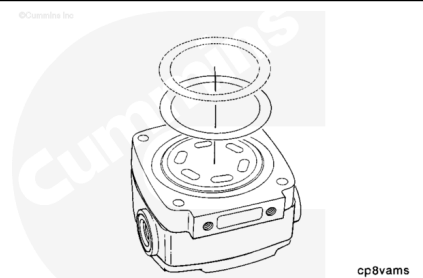
Remove the head, cover, and valve plate assembly and place it on a clean work surface with the intake valve facing upward.



cp8hdmg

Some units have a press-fit intake valve retainer. If present, carefully remove it to prevent part damage.

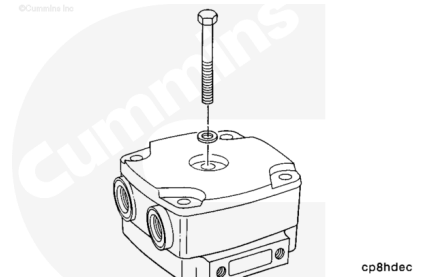
Remove the intake valve.



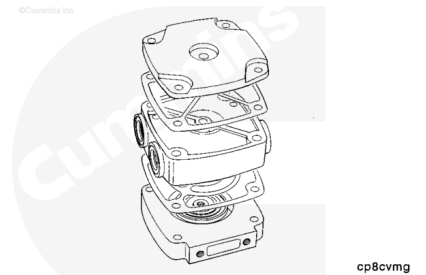
cp8vams

Note : The center capscrew is shorter than the four corner capscrews.

Turn the head assembly over and set it on a clean surface. Remove the center capscrew and the capscrew can be used again.

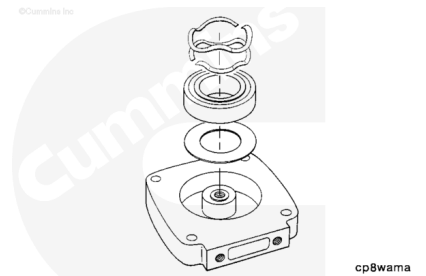


Remove the cover, cover gasket, head, and head gasket.



Note : The wave washers must be replaced.

Remove the two wave washers, exhaust valve retainer, and exhaust valve.

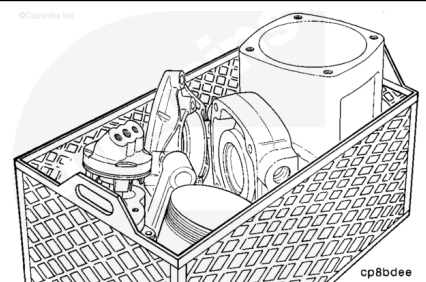


Clean and Inspect for Reuse

⚠ WARNING ⚠

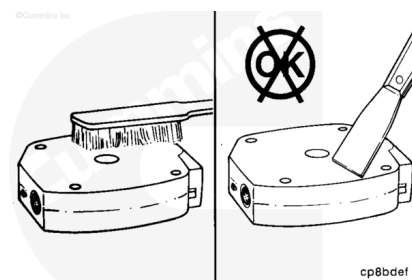
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing.

Soak the parts in a kerosene emulsion-based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier can be contacted about solution concentration, temperature, and soak time.

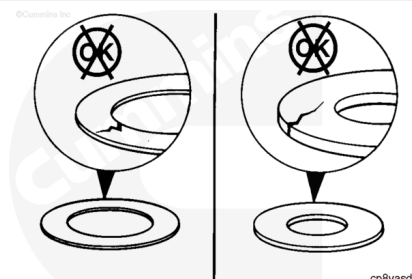


Note : Do not use a scraper to remove carbon and scale, the sealing surfaces can be damaged.

The parts can be scrubbed with a stiff nonmetallic bristle brush.



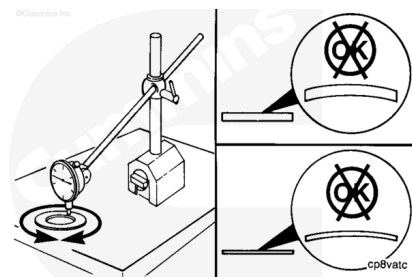
Inspect the intake and exhaust valves for cracks or damage.



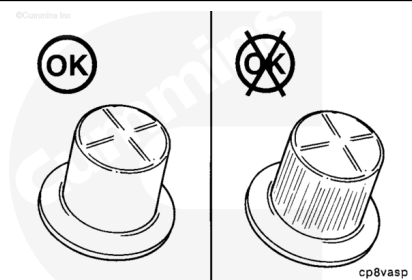
Note : Cummins Inc., recommends new valves be installed.

Measure the flatness of the intake and exhaust valves. Both valves **must** be flat within 0.03 mm [0.001-inch].

Replace valves if cracked, damaged, or **not** flat.



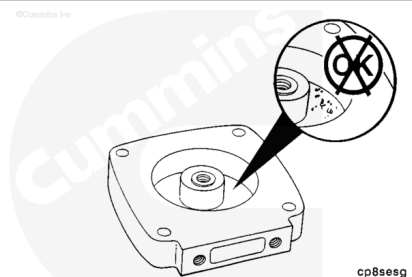
Inspect the upper part of the unloader valve cap where the rectangular V-seal operates. Check for scoring.



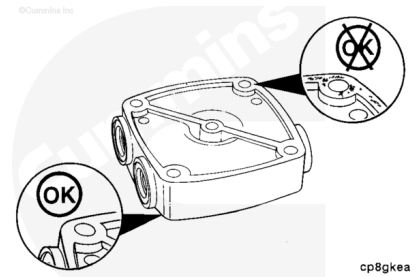
Note : Inspection of the valve seats in the valve plate requires specialized equipment and is beyond the scope of field service.

Inspect the valve seat surfaces.

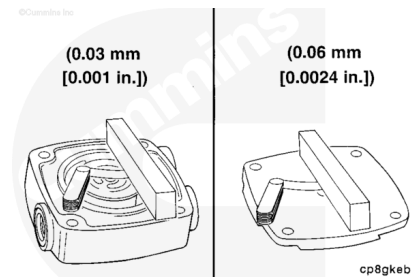
If the valve seat is damaged, or can **not** be cleaned, a new valve plate is available in a service kit. Otherwise, a QE valve plate service assembly can be used.



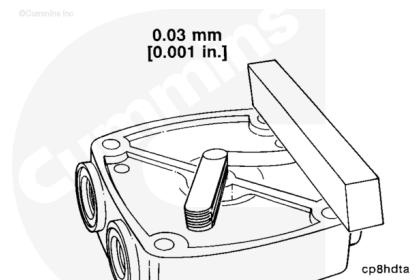
Gasket sealing surfaces **must** be clean and free of all old gasket material, carbon, rust, and other buildup. Surfaces **must** be free of scratches, gouges, burrs, and other deformities.



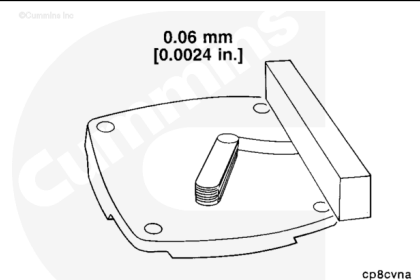
After making sure all gasket surfaces are clean and free of the above, inspect the head and cover for flatness. Use the flat plate and the feeler gauges.



Single cylinder heads **must** be flat within 0.03 mm [0.001-in] between any two adjacent capscrew holes.



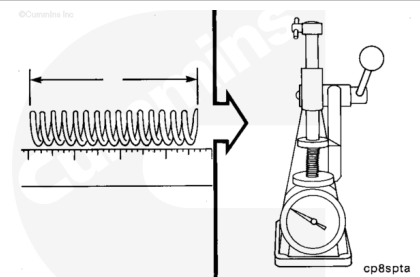
Single cylinder top cover **must** be flat within 0.06 mm [0.0024-in] between any two adjacent capscrew holes and 0.10 mm [0.004-inch] total.



Note : Cummins Inc., recommends that new springs be installed.

Use valve spring tester, Part Number 3375182, or equivalent, to check the unloader valve spring. Replace any spring that does **not** meet specifications.

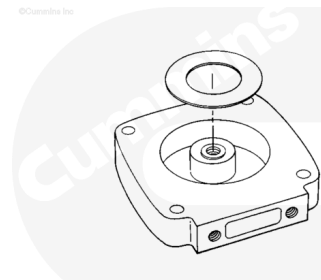
Reference the Compressor Spring Specifications chart in Section V.



Assemble

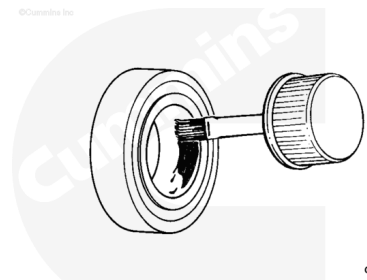
Holset® Non-European A/C Model

Install the exhaust valve over the post in the valve plate.



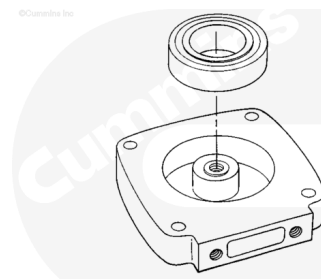
cp8vahk

Apply a thin coating of antiseize to the inside diameter of the exhaust valve retainer.



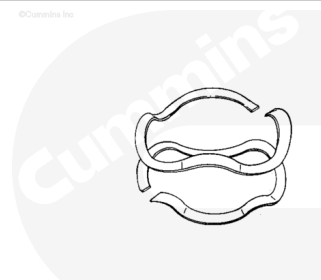
cp8rtwa

Slide the exhaust valve retainer over the valve plate. Make sure that the end of the retainer with the groove faces upward.



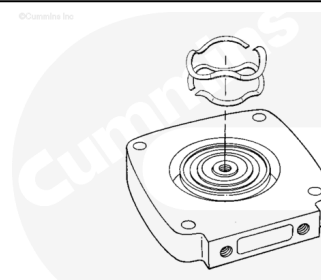
cp8rtha

Align the wave spring gaps 180 degrees from each other **not** to overlap.



cp8sphe

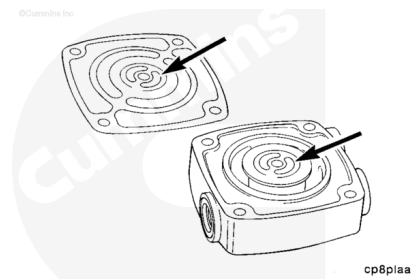
Place the wave springs in the retainer groove.



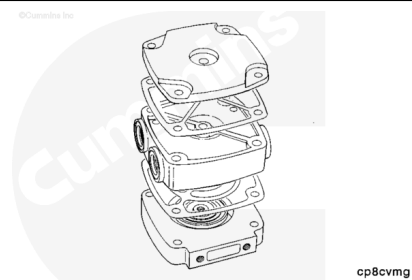
cp8spht

Determine the final orientation of the valve plate (air intake location) and the head (coolant ports with respect to air inlet or manifold location). Align the kidney-shaped slots in the head with the kidney shaped slots in the gasket.

If orientation marks were made before disassembly, use them.



Assemble the cover, cover gasket, head, head gasket, and valve plate.

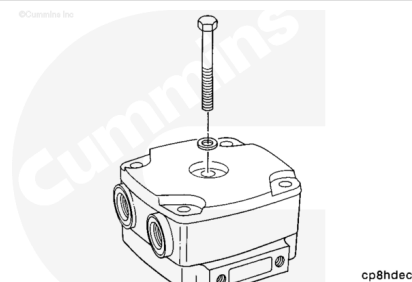


Note : Make sure corner capscrew holes are aligned.

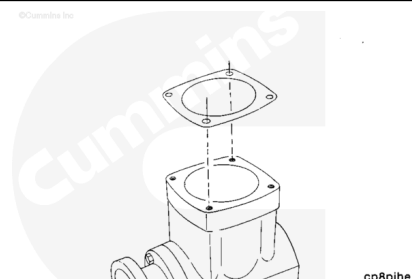
Lubricate the threads under the head.

Install the shorter capscrew with washer through the center hole.

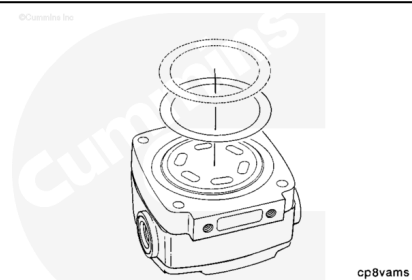
Torque Value: 14 n•m [10 ft-lb]



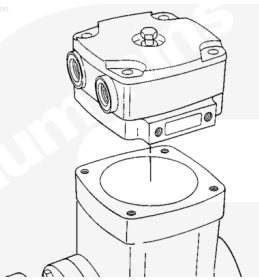
Install the valve plate gasket.



Carefully place the intake valve in the valve plate. Install the intake valve retainer.



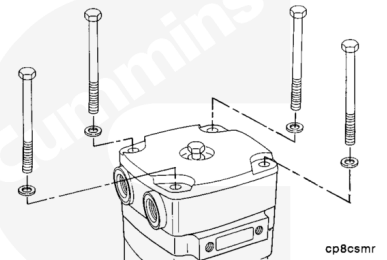
Install the valve plate assembly.



cp8hdmg

Lubricate the threads under the head and washer of the capscrews if initially installed.

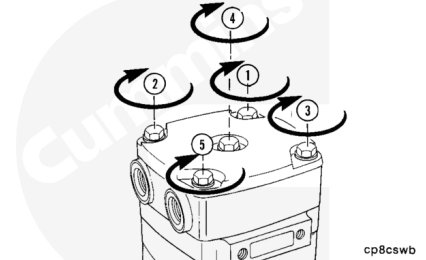
Install the four head capscrews and washers.



cp8csmr

Tighten all five capscrews.

Torque Value: 28 n·m [21 ft-lb]

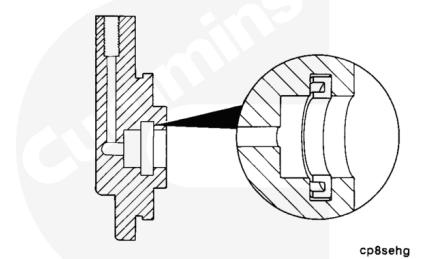


cp8cswb

Note : The seal must be installed with the grooved side up.

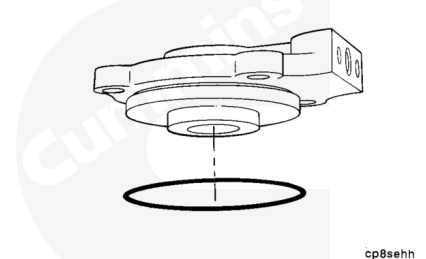
Install the new rectangular V-seal into the unloader body.

Liberal lubricate the unloader valve bore above and below the rectangular ring seal with high temperature Accrolube Lubrication Teflon grease, or equivalent.



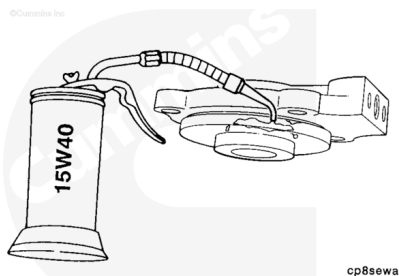
cp8sehg

Install a new o-ring seal on the unloader valve body.



cp8sehh

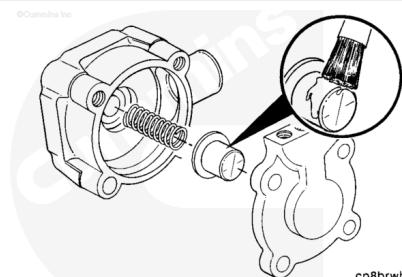
Use clean 15W-40 oil or Accrolube Lubrication Teflon Grease, or equivalent, to lubricate the seal.



Liberalily lubricate the unloader valve body bore and unloader cap with high temperature Accrolube Lubrication Teflon grease, or equivalent.

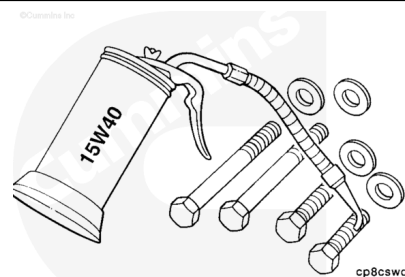
Install the unloader cap.

Install the unloader spring.



Note : The two unloader body screws must not be used to attach any brackets.

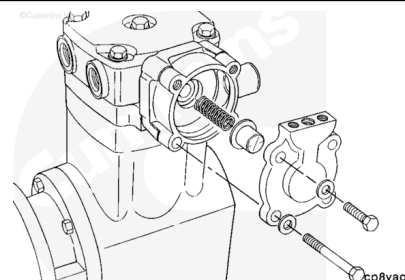
Lubricate the unloader screw threads and under head with clean engine oil (SAE 15W-40), before installation.



Note : The longer capscrews are used to mount the manifold to the air compressor.

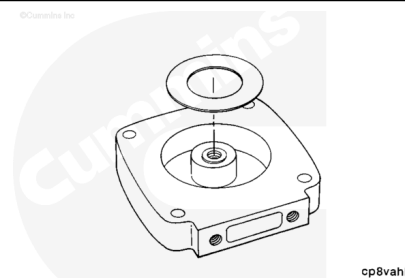
Assemble the unloader components and attach the unloader assembly to the valve plate with the four capscrews and washers.

Torque Value: 27 n•m [20 ft-lb]

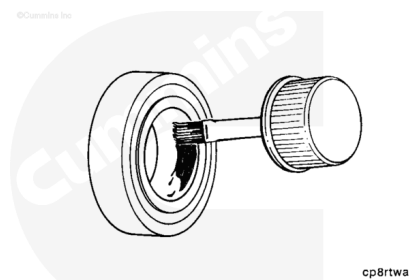


Holset® European A/C Model

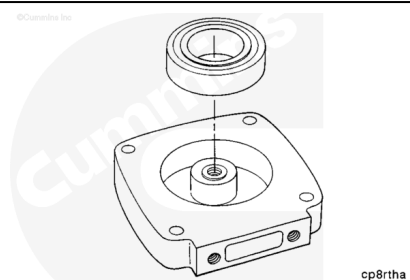
Install the exhaust valve over the post in the valve plate.



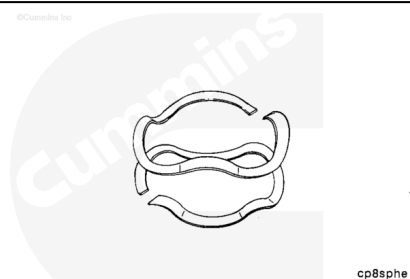
Apply a thin coating of antiseize to the inside diameter of the exhaust valve retainer.



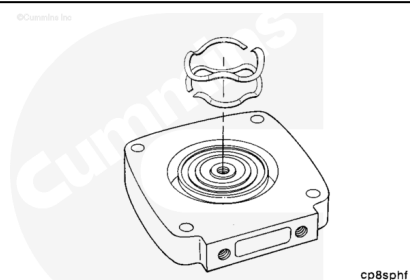
Slide the exhaust valve retainer over the valve plate. Make sure that the end of the retainer with the groove faces upward.



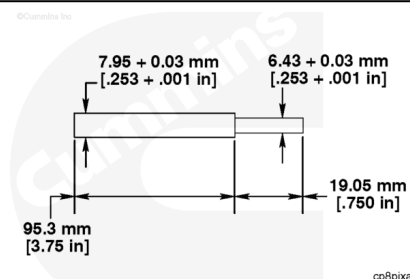
Align the wave spring gaps 180 degrees from each other so they do **not** overlap.



Place the wave springs in the retainer groove.

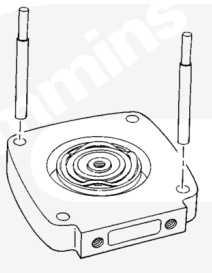


Fabricate or reuse the four guide pin tools.



Insert the larger end of two guide pins in opposite corner holes of the valve plate.

©Cummins Inc.

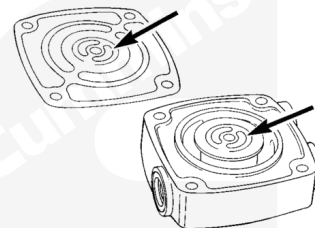


cp8pihc

Determine the final orientation of the valve plate (air intake location) and the head (coolant ports with respect to air inlet or manifold location). Align the kidney-shaped slots in the head with the kidney shaped slots in the gasket.

If orientation marks were made before disassembly, use them.

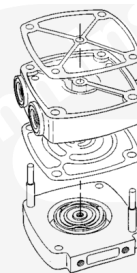
©Cummins Inc.



cp8piaa

Install the head gasket onto the guide pins (either side up, but with correct slot orientation). Install the head onto the guide pins with the kidney-shaped slots aligned and toward the valve plate. Install the cover gasket over the guide pins.

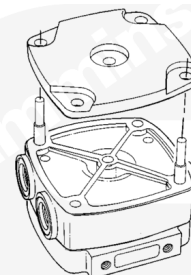
©Cummins Inc.



cp8gkhf

Assemble the cover.

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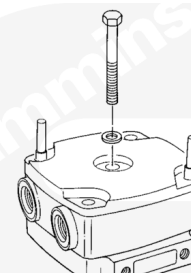


cp8cvhb

Install the shorter capscrew with washer, if initially installed, through the center hole.

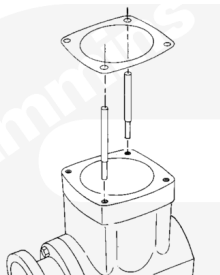
Torque Value: 14 n•m [120 in-lb]

©Cummins Inc.



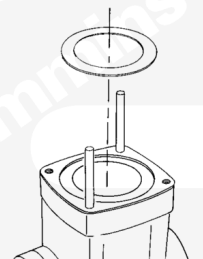
cp8cshh

Place the remaining two guide pins in the crankcase head capscrew holes (that will **not** interfere with the guide pins already in the head assembly). Install the valve plate gasket.



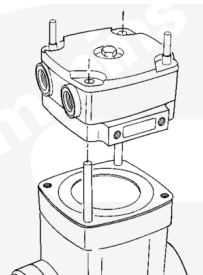
cp8pihd

Carefully place the intake valve on the crankcase, located by the valve plate gasket. Do **not** allow the valve to overlap the gasket.



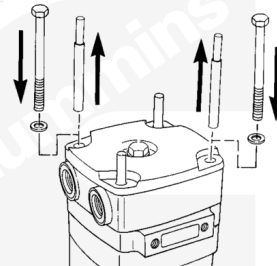
cp8vah1

Install the head assembly over the guide pins. Be careful **not** to disturb the location of the intake valve. The compressor will **not** work if the valve overlaps the gasket and is pinched.



cp8hdhb

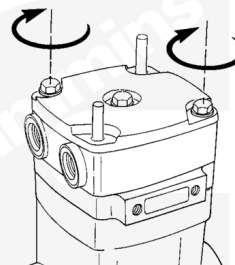
Carefully remove two of the guide pins and replace with two head capscrews and washers, if initially installed.



cp8pimb

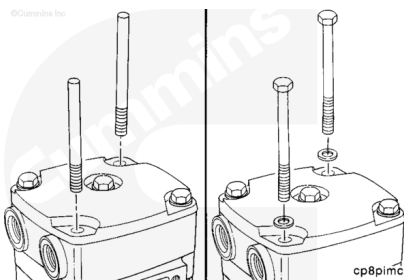
Tighten the two head capscrews.

Torque Value: 14 n•m [10 ft-lb]



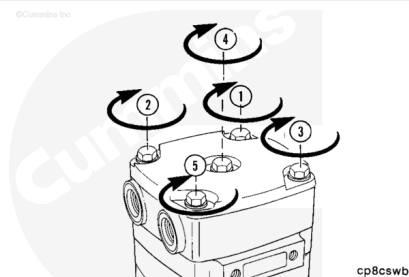
cp8cswa

Remove the remaining two pins and replace with two head capscrews and washers, if initially installed.



Tighten all five capscrews.

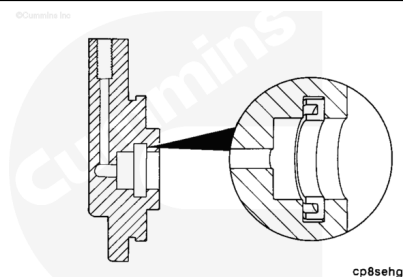
Torque Value: 28 n•m [21 ft-lb]



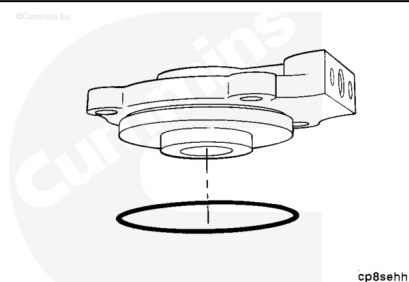
Note : The seal must be installed with the grooved side up.

Install the new rectangular V-seal into the unloader body.

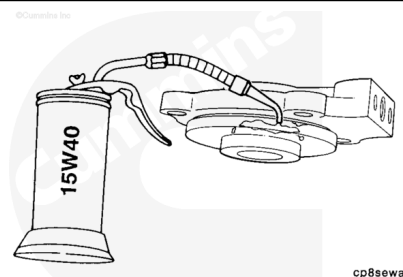
Liberalily lubricate the unloader valve bore above and below the rectangular ring seal with high temperature Accrolube Lubrication Teflon grease, or equivalent.



Install a new o-ring seal on the unloader valve body.



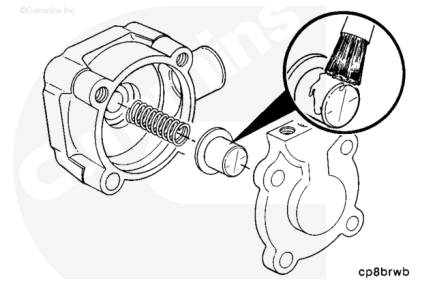
Use clean 15W-40 oil or Accrolube Lubrication Teflon Grease (or equivalent) to lubricate the seal.



Liberalily lubricate the unloader valve body bore and unloader cap with high temperature Accrolube Lubrication Teflon grease, or equivalent.

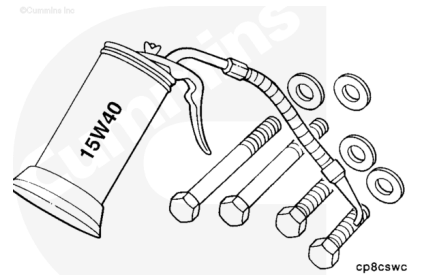
Install the unloader cap.

Install the unloader spring.



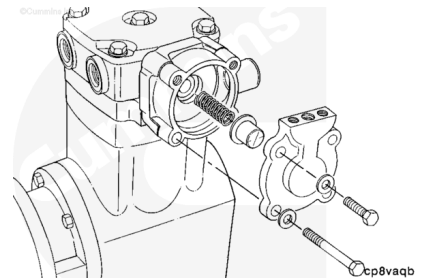
Note : The two unloader body screws must not be used to attach any brackets.

Lubricate the unloader screw threads and under the head with clean engine oil (SAE 15W-40) before installation.



Note : The longer capscrews are used to mount the manifold to the air compressor.

Assemble the unloader components and attach the unloader assembly to the valve plate with the four capscrews and washers.



Torque Value: 27 n•m [20 ft-lb]